

Training Convolutional Networks with Spatial Continuation

Subject 24 – Homotopy-Based Training of Deep Neural Networks

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Outline

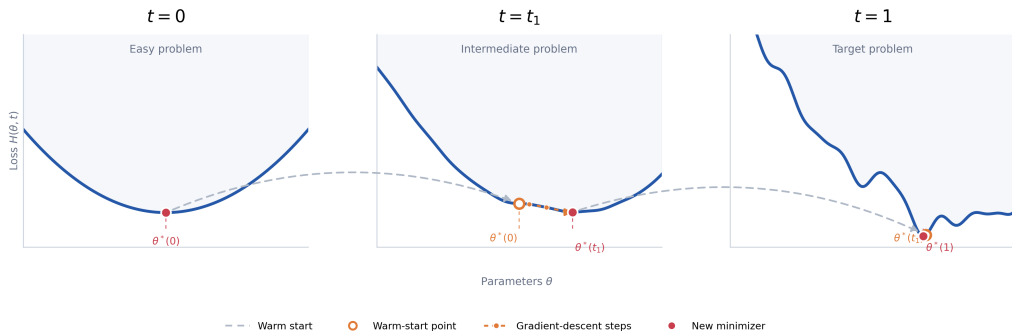
- 1 Motivation
- 2 Spatial continuation
- 3 Selection on CIFAR-10
- 4 Benchmarks
- 5 Loss landscape
- 6 Conclusion

Homotopy-Based Training of Deep Neural Networks

Training deep neural networks involves navigating high-dimensional, non-convex loss landscapes.

Homotopy idea: start with an easier problem and progressively transform it into the target problem as t goes from 0 to 1:

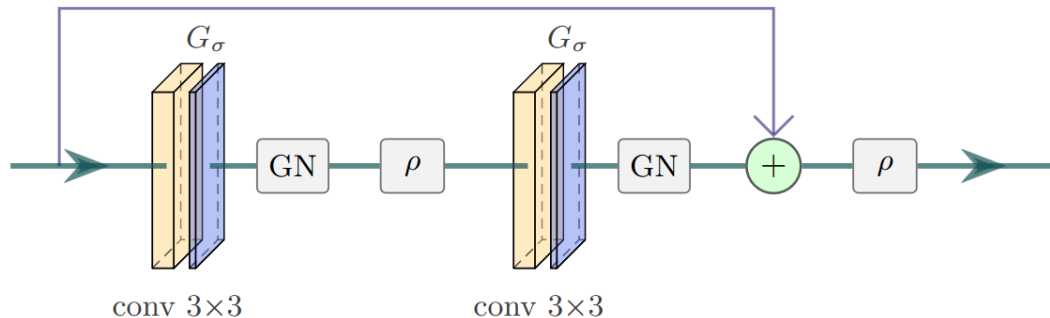
$$H(\theta, 0) = g(\theta) \quad \longrightarrow \quad H(\theta, 1) = f(\theta).$$



In our setting: the progression acts on the spatial information processed by the model during training.

Inspiration from the literature

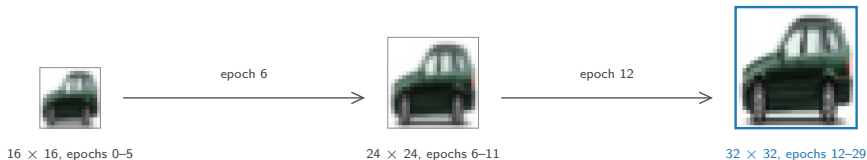
Our main inspiration: Curriculum by Smoothing [11]



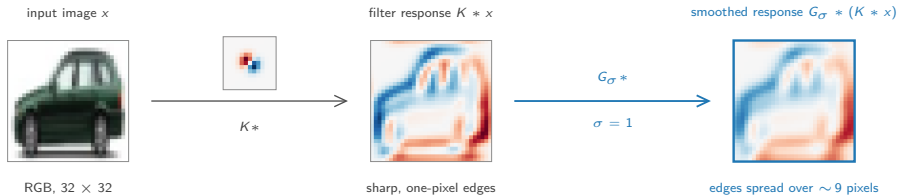
- Gaussian kernel G_σ applied to the feature maps
- σ decreases during training, original network at the end

Feature Map Smoothing and Progressive Resolution

Progressive feature-map resolution: $16 \rightarrow 24 \rightarrow 32$



Decreasing Gaussian smoothing of a convolutional feature map



Our methodology to retain 3 approaches

Exploration on CIFAR-10 / ResNet-20

- ① Blur the input images
→ worse: 71.3% vs 84.1% ($\sigma = 1$)
- ② Blur the feature maps
→ better: 78.4% vs 75.4%
- ③ Process smaller feature maps
→ 29 configurations, 87 runs: operators, locations, schedules
- ④ Final paired batch
→ 16 configurations $\times$ 3 seeds $\Rightarrow$ **R**, **G**, **RG**

Discarded

- Gaussian with GroupNorm (-3.5 pp)
- Perceptual pooling (diverged)
- db2 wavelets ($43\times$ slower)
- Low resolution until the end (73–76%)
- Per-site σ profiles (no consistent gain)

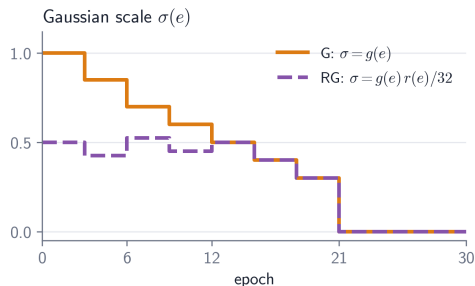
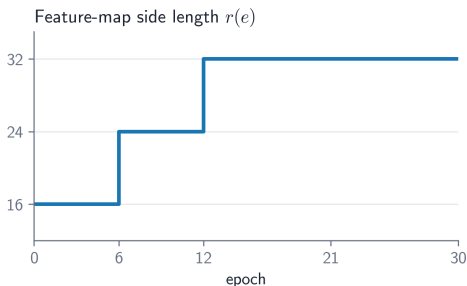
Not better than max-pooling

SoftPool, bilinear, L^2 and $\dot{H}^{-1}$ reconstructions

Pilots used other splits and budgets; selection reused the CIFAR-10 test set.

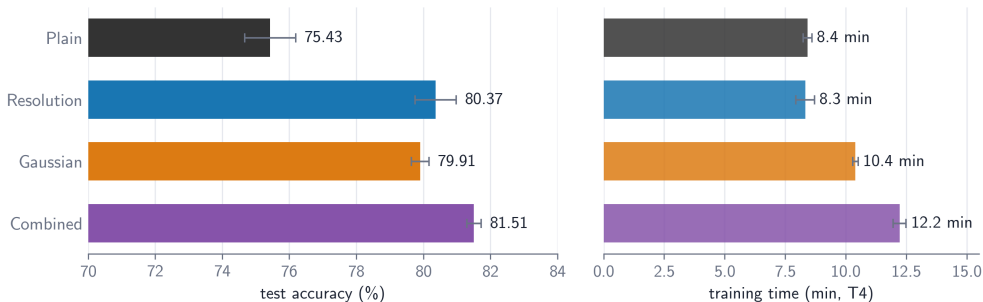
The three retained procedures

Method	Where	Schedule
R Resolution	adaptive max-pooling after block 1	$16 \rightarrow 24 \rightarrow 32$, full size from epoch 12
G Gaussian	10 post-ReLU sites	σ : $1 \rightarrow 0.3$, removed from epoch 21
RG Combined	R + Gaussian at the 19 conv outputs	σ scaled by $r(e)/32$



- All three end at the original network: no cost at inference

Selection batch



Resolution

+4.9 pp

Gaussian

+4.5 pp

Combined

+6.1 pp

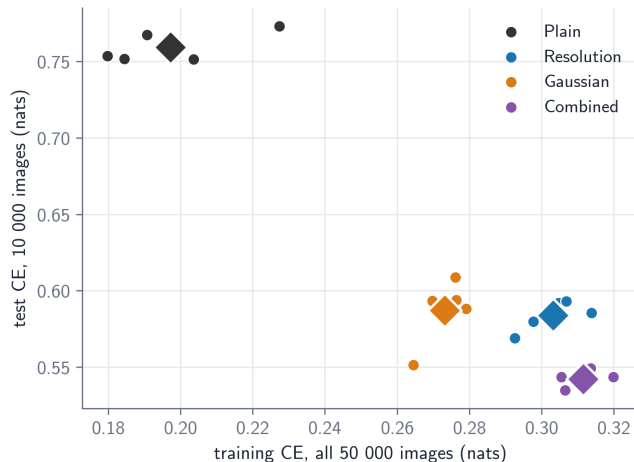
CIFAR-10 (50k), ResNet-20-BN, SGD, 30 epochs, Tesla T4. Mean $\pm$ SD over 3 seeds; gains paired within seed.

Our final benchmarks

Dataset	Model / norm / activation	Optimizer	Epochs	Plain	R	G	RG
CIFAR-10	ResNet-20 / BN / ReLU	SGD	30	75.4 $\pm$ 0.8	80.4 $\pm$ 0.6	79.9 $\pm$ 0.3	81.5 $\pm$ 0.2
CIFAR-10	ResNet-20 / BN / ReLU	Adam	30	84.1 $\pm$ 0.6	85.3 $\pm$ 0.3	84.0 $\pm$ 0.1	85.3 $\pm$ 0.3
CIFAR-10	ResNet-20 / BN / ReLU	AdamW	30	84.1 $\pm$ 0.5	86.3 $\pm$ 0.2	83.7 $\pm$ 0.3	86.2 $\pm$ 0.3
CIFAR-10	ResNet-20 / BN / ReLU	RAdam	30	84.4 $\pm$ 0.1	85.6 $\pm$ 0.4	84.7 $\pm$ 0.2	85.3 $\pm$ 0.2
CIFAR-10	ResNet-20 / BN / GELU	SGD	30	74.8 $\pm$ 0.6	80.8 $\pm$ 0.4	80.3 $\pm$ 0.4	81.6 $\pm$ 0.1
CIFAR-10	ResNet-20 / BN / SiLU	SGD	30	76.5 $\pm$ 0.7	80.9 $\pm$ 0.3	80.4 $\pm$ 0.8	81.7 $\pm$ 0.1
CIFAR-10	ResNet-20 / GN / ReLU	SGD	30	75.5 $\pm$ 0.9	78.3 $\pm$ 0.7	76.6 $\pm$ 0.4	78.9 $\pm$ 0.2
CIFAR-10	VGG-11 / BN / ReLU	SGD	30	80.7 $\pm$ 0.6	83.8 $\pm$ 0.4	84.7 $\pm$ 0.2	84.5 $\pm$ 0.5
SVHN	ResNet-20 / BN / ReLU	SGD	30	93.1 $<$ 0.1	94.3 $\pm$ 0.1	94.3 $\pm$ 0.1	94.8 $\pm$ 0.2
STL-10	ResNet-20 / BN / ReLU	SGD	60	57.3 $\pm$ 0.5	59.9 $\pm$ 0.4	58.4 $\pm$ 1.5	56.1 $\pm$ 0.4
CIFAR-10 (5k)	ResNet-20 / BN / ReLU	SGD	60	48.8 $\pm$ 1.1	55.0 $\pm$ 0.5	57.4 $\pm$ 0.4	59.4 $\pm$ 1.3
CIFAR-10 (5k)	ResNet-20 / BN / ReLU	SGD	293	50.8 $\pm$ 0.9	55.0 $\pm$ 0.8	60.9 $\pm$ 0.2	62.5 $\pm$ 0.9

Final test accuracy (%), mean $\pm$ SD over 3 seeds, no augmentation. **Bold**: best in row. **Red**: below Plain. Both 5k rows use the same 5 000 CIFAR-10 images.

Higher training loss, lower test loss

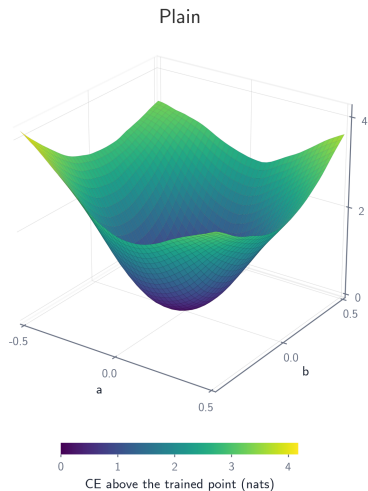


- Training loss $\uparrow$, test loss $\downarrow$, accuracy $\uparrow$
- All three methods, in 5 / 5 seeds
- Same pattern in 22 / 26 transfer comparisons

Full training and test sets, saved BatchNorm statistics.

Diamonds: means over 5 seeds.

Visualization of the loss landscape (1): Plain



Height at (a, b) :

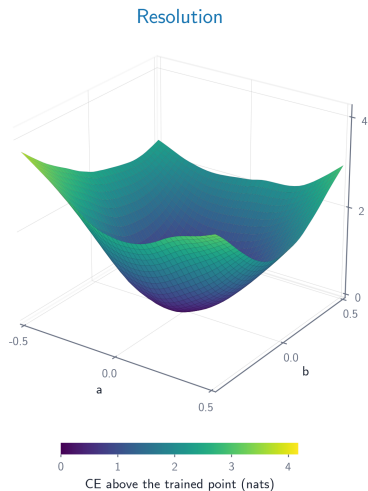
$$L(\theta + a d_1 + b d_2) - L(\theta)$$

Mean rise over the plane

1.92 nats

CIFAR-10, ResNet-20-BN, SGD, final model, seed 0.

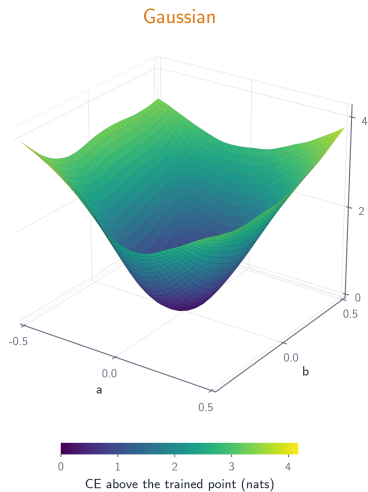
Visualization of the loss landscape (2): Resolution



Mean rise over the plane

1.55 nats

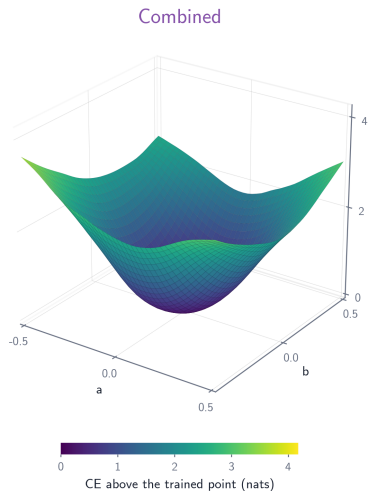
Visualization of the loss landscape (3): Gaussian



Mean rise over the plane

2.05 nats

Visualization of the loss landscape (4): Combined

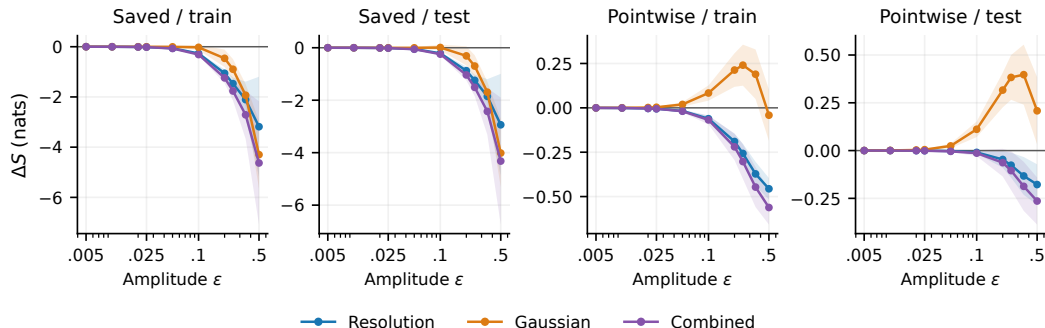


Mean rise over the plane

1.53 nats

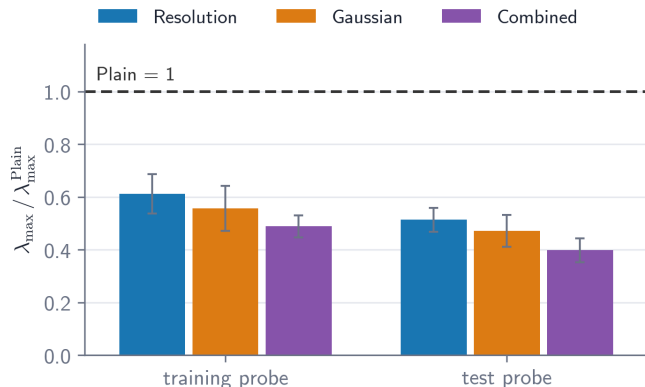
Perturbation sensitivity depends on BatchNorm

$$S(\varepsilon) = \frac{1}{K} \sum_{k=1}^K \left[\frac{L(\theta + \varepsilon d_k) + L(\theta - \varepsilon d_k)}{2} - L(\theta) \right], \quad \Delta S = S_{\text{method}} - S_{\text{Plain}}$$



- Saved statistics: all three flatter than Plain at large ε
- Recalibrated statistics: **G** sharper than Plain on the test probe

Curvature: largest Hessian eigenvalue



Relative to Plain

0.4 – 0.6×

- All three methods, 5 / 5 seeds, both probes
- Lanczos on Hessian–vector products, 268 336 weights, frozen BatchNorm
- Negative eigenvalues too: not certified minima

Takeaways

Shrinking feature maps early helps

12 / 12 settings, about the same training time, nothing changes at inference

The Gaussian is recipe-dependent

Helps with SGD, slightly negative with Adam(W), the combination hurts on STL-10

Flatness is not the whole story

Lower curvature for all three methods, but the sensitivity ordering flips with BatchNorm

Limits

- Selection reused the CIFAR-10 test set
- No data augmentation
- 3 to 5 seeds per setting
- Surfaces: one illustrative plane

Next steps

- Compare with CBS and SDPoint
- Theory: higher train loss, lower test loss
- Larger models and datasets
- Landscapes around the transitions

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